Krishnaa **Vadivel** | Software Engineer | Scientific Computing | HPC Workflows

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Profile

Software engineer and PhD researcher focused on scientific computing, distributed workflows, numerical software, and engineering tooling. Strong background in Python and C++ development for simulation, automation, data processing, visualization, and infrastructure spanning research and production-adjacent environments.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*

- Develop scientific software for first-principles and many-body workflows in materials research, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, and Linux-based automation.
- Build distributed and accelerator-enabled workflows for HPC environments including Frontier, Frontera, and Perlmutter, with experience spanning CUDA-, ROCm-, and container-oriented development patterns.
- Create reusable tooling, notes, and structured workflows that bridge theory, implementation, and reproducibility in computational science.
- Extend into ML-assisted and retrieval-driven automation where useful, particularly for scientific-data generation and research-workflow setup.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA-accelerated processing and custom OpenGL visualization software for large acoustic waveform datasets used in engineering diagnostics and field-tool analysis.
- Developed ASP-style services and Microsoft SQL-backed workflows for internal software and field-data management.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.
- Participated in broader engineering support spanning embedded-adjacent software, validation, and integration contexts.

FindLight
Photonics Technical Writer *2018–2019*

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

Selected Projects

DOLFINx Semiconductor and PDE Workflow

- Built Python workflows around DOLFINx and PETSc for Poisson and drift-diffusion style simulation, including theory notes, implementation scaffolding, and numerical experimentation.

Agentic Scientific Workflow Generation

- Developed retrieval-driven tooling that reads documentation and academic papers to generate structured starter inputs and workflow artifacts for computational-science tasks.

Scientific Visualization and GPU Tooling

- Built CUDA/OpenGL-based tooling and related utilities for large numerical datasets, interactive inspection, and engineering communication.

Photonics and Simulation Code Artifacts

- Created clean, inspectable software artifacts for FDTD-based photonics simulation and related numerical-science workflows.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Selected Coursework

Computing Foundations: Theoretical Computer Science; Probabilistic Machine Learning

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, PowerShell, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux command line, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, OpenGL, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, scikit-learn, agentic workflows, retrieval-driven automation